VENDOR RECONCILE — Detailed Enterprise Blueprint

Comprehensive SAP-Connected Reconciliation & Match Engine Architecture

COMPLETE SOLUTION OVERVIEW

Version 3.0 • September 2026

Executive Summary

Vendor Reconcile is a robust, enterprise-grade execution platform built on the **SAP Cloud Application Programming (CAP)** framework and Node.js. It is specifically engineered to eliminate manual financial discrepancies by acting as a highly automated verification layer between internal procurement workflows and external vendor billing.

The system has achieved complete live integration with the **SAP S/4HANA (S21) ERP** system, actively pulling Purchase Orders, Goods Receipts, and Vendor Master Data natively via secure OData APIs. By orchestrating a mathematically rigorous 3-Way Matching Engine, the platform empowers Accounts Payable teams to achieve touchless processing for clear invoices while providing a dedicated dispute management portal for exceptions.

1. CORE PROJECT PILLARS & VALUE PROPOSITION

✓ End-to-End P2P Governance

Complete lifecycle management from Purchase Requisition creation to UTR payment clearing, governed by strict Role-Based Access Control (RBAC) across five distinct operational personas.

✓ Automated Invoice Matching

System-driven 3-way matching logic automatically simulates MIRO clearance, highlighting exact price/quantity variances down to the decimal without manual cross-referencing.

✓ Seamless SAP S21 Sync

Bi-directional data continuity via secure REST/OData APIs ensures the platform always acts on a unified operational truth, eliminating data silos between logistics and finance.

2. DEEP-DIVE ARCHITECTURE & TECHNOLOGY STACK

The application operates on a modern, decoupled architecture ensuring massive scalability, secure data custody, and responsive user experiences.

3. PRESENTATION LAYER (SAP Fiori Elements / UI5)

- Executive AP Dashboard
- Data Intake & Lines Workspace
- Vendor Exception Portal

⇅ **Secure OData V4 / REST Endpoints** ⇅

2. OPERATIONAL CORE PLATFORM (Node.js & SAP CAP Framework)

- Automated 3-Way Matching Engine
- KPI Aggregation Service
- Exception Workflow Manager
- S21 API Sync Connector

⇅ OData V2 Basic Auth Proxy ⇅

1. ENTERPRISE SYSTEM OF RECORD (SAP ECC / S/4HANA S21)

Procurement (MM)	Inventory Mgmt (MIGO)	Financial Acct (FI)	Vendor Master Data

Backend Engineering

- **Framework:** Node.js with SAP Cloud Application Programming (CAP) Model v4.
- **Database:** High-performance transactional SQLite (Local) engineered for seamless transition to SAP HANA Cloud.
- **Service Definitions:** Domain-driven CDS (Core Data Services) schemas modeling MatchResults, Exceptions, and VendorLedgers.

Frontend Engineering

- **UI Framework:** SAP UI5 / Fiori Elements.
- **Data Binding:** Dynamic JSON and OData models mapping directly to backend endpoints.
- **Analytics:** Microcharts, dynamic formatters, and KPI visualizers natively rendering real-time matching outcomes.

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3. LIVE SAP INTEGRATION MAPPING (S21 DATA FLOW)

The application is rigorously connected to the live SAP S21 ERP system. The custom `s21-connector.js` orchestrates synchronization, guaranteeing that local data precisely mirrors SAP business reality.

SAP MODULE	INTEGRATION ENDPOINT	ENGINEERED IMPLEMENTATION DETAILS
Purchase Orders (ME23N)	`API_PURCHASEORDER_PROCESS_SRV/A_PurchaseOrder`	Extracts expected pricing, delivery terms, and PO line items. **Recent Upgrade:** Dynamically parses the PO payload to extract the true `AddressName` for suppliers, replacing numeric ID fallbacks.
Supplier Invoices (MIRO)	`API_SUPPLIERINVOICE_PROCESS_SRV/A_SupplierInvoice`	Ingests supplier billing data. **Recent Upgrade:** Since the SAP invoice item endpoint lacks material codes, the engine now actively cross-references the PO database to map true material IDs (e.g., 10314) to invoice lines.
Goods Receipts (MIGO)	`API_MATERIAL_DOCUMENT_SRV/A_MaterialDocumentItem`	Extracts physical warehouse intake quantities, allowing the engine to calculate whether a vendor is billing for unreceived freight (Uninvoiced GR tracking).

4. THE AUTOMATED 3-WAY MATCHING ENGINE

At the heart of the platform is the custom Node.js verification engine (`matching-engine.js`). It evaluates aggregated SAP data against vendor uploads to determine financial clearance.

✓ CLEARED MATCHED / 3-WAY-CLEAR

Logic: The system successfully locates the PO in SAP, verifies the invoiced price is within 2% tolerance of the expected PO rate, and ensures the invoiced quantity does not exceed the warehouse GR logged quantity.

Action: Approved for automated MIRO processing and UTR payment generation.

⚠ DISPUTE UNMATCHED / PRICE-DIFF

Logic: The expected PO price differs from the Vendor's billed amount by more than the strict organizational threshold.

Action: Automatically generates an Exception ticket allocated to the Finance Exec queue.

⚠ DISPUTE UNMATCHED / QTY-EXCEEDS-GR

Logic: The vendor is attempting to bill for a higher volume of material than what was physically received and scanned by the GR Specialist.

Action: Routed to Procurement for vendor credit-note requests.

✕ CRITICAL UNMATCHED / PO-NOT-FOUND

Logic: A vendor invoice was uploaded, but the engine cannot locate any active Purchase Order reference within the synced SAP database.

Action: Immediate AP Manager intervention required to block fraudulent billing.

5. EXECUTIVE DASHBOARD & KPI INTELLIGENCE

The dashboard (`Cockpit.controller.js` & `kpi-handler.js`) mathematically aggregates the outcomes of the matching engine in real-time, yielding critical operational insights.

Missing in File

Actively queries the `GrirSnapshot` to locate open SAP Goods Receipts that have no corresponding vendor invoice attached to them yet.

Duplicates

Scans `VendorInvoiceHeaders` dynamically to catch overlapping, identical invoice submissions from a single vendor to prevent double-billing.

High-Value Exceptions

Continuously monitors `MatchResults` to flag any singular invoice dispute where the financial variance exceeds ₹50,000.

Vendor Risk Mapping

Compiles a live ledger ranking vendors by the volume of their active disputes (e.g., Endurance Components, Stark Industry).

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6. ROLE-BASED OPERATIONAL WORKFLOWS

The system enforces strict RBAC (Role-Based Access Control) to govern the reconciliation journey. Distinct personas operate within their isolated UI scopes.

PR

PR Indenter

Initiates the procurement cycle by raising Purchase Requisitions for required raw materials or services within SAP.

PO

PO Officer

Audits requisitions, approves pricing, converts PRs to official Purchase Orders, and dispatches them to external suppliers.

VEN

Vendor

Logs into a restricted self-service portal to view awarded POs, mark dispatch statuses, and upload digital invoice copies.

GR

GR Specialist

Manages physical logistics, recording MIGO goods receipts directly reflecting actual warehouse intake.

FIN

Finance Exec

Reviews matching engine exceptions, negotiates disputes, overrides tolerances, and clears invoices for UTR execution.

7. DATA GOVERNANCE & SCHEMA ARCHITECTURE

To isolate transactional telemetry without polluting the SAP Master Data environment, the CAP Application utilizes a structured internal database schema comprising over 10 interconnected entities.

CORE ENTITY (DB)	PURPOSE & RELATIONAL MAPPING
`VendorMaster`	Stores normalized supplier profiles. Synced dynamically via SAP PO endpoints to capture accurate `businessPartnerName`, avoiding raw numeric ID leakage in UI displays.
`PurchaseOrders`	Mirrors active SAP ME23N records. Contains granular line-item details (Qty, Amount, Material ID) serving as the baseline truth for the Matching Engine.
`VendorInvoiceHeaders` & `Items`	Houses ingested supplier billing documents (either synced from SAP MIRO or uploaded via portal). Line items cross-reference `PurchaseOrders` to derive true expected materials.
`MatchResults`	The immutable audit ledger written exclusively by the Matching Engine. Links an Invoice Item to a PO Item and statically records the evaluated `outcomeCode`, absolute `amountDelta`, and rule applied.
`Exceptions`	Spawns conditionally when a `MatchResult` fails verification (e.g. `PRICE-DIFF`). Tracks resolution status (OPEN/RESOLVED), SLA aging, and assigns priority (P1/P2/P3) for Finance AP queues.

8. STRATEGIC ROADMAP & NEXT PHASES

1. Native OCR Ingestion Pipeline

Integration with Microsoft/Grok optical character recognition to fully automate the Data Intake process. Unstructured vendor email PDFs will be autonomously scraped, validated,

2. Automated UTR Settlement (Outbound)

Enhancing the outbound proxy connectors to actively push `MATCHED/3-WAY-CLEAR` invoices back into SAP F110 (Automatic Payment Run). This will close the loop,

and inserted into the `VendorInvoiceHeaders` table without human keystrokes.

automatically clearing payables and echoing the final Bank UTR numbers back to the Vendor UI.